

Testing VIVACE efficiency

An experimental investigation on the effect of incident fluid flow rate and system mass on the oscillation efficiency of a VIVACE generator.

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Abstract

A large portion of the available clean energy in ocean and river currents remains unharvested as it is outside the range of financially viable flow rates for large scale hydro turbines. Vortex Induced Vibrations for Aquatic Clean Energy (VIVACE) aims to generate hydro-kinetic energy from lower flow rates, such as in rivers, propelled by vortex shedding behind a cylindrical bluff body in fluid flow. In this experiment a VIVACE system was designed and manufactured. The spring constant of the system was determined using Hooke's law to be $25.26 \pm 0.01 \text{ Nm}^{-1}$, and the damping coefficient of the system was found to be $0.194 \pm 0.001 \text{ Nsm}^{-1}$ by monitoring the unforced lightly damped motion of the system. These two values were used in the calculation of VIVACE generation efficiencies, by plotting and fitting data collected through applying motion tracking software to footage of the converter. The system achieved peak efficiencies between 0.096 ± 0.022 at the lowest mass of $m = 263.1 \text{ g}$ and 0.215 ± 0.022 at the largest mass of $m = 301.8 \text{ g}$. The greatest of which is within the range previously collected by Sun et al. (2016), and is 70% of that achieved by Bernitsas et al. (2009). It was found that the relationship between flow velocity and system efficiency is Gaussian due to the resonant nature of the oscillator, and that the relationship between oscillator mass and efficiency is linear for small masses.

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1 Introduction

Reports investigating the utilisation of marine currents as a power source such as Hall (2006), World Energy Council (2001) and Pontes & Falcao (2001) have suggested that a substantial portion of the world's energy usage could be satisfied by hydrokinetic generation. Currently the majority of energy generated from oceans is via turbines, which Previsic & Bedard (2005) suggests are only financially viable at average flow rates within 2.5 - 3.6 m/s (5 - 7 knots). This presents a problem, as a large portion of the worlds currents posses flow speeds below 3 knots, with rivers typically flowing as slowly 2 knots (1.03 m/s) (Lee & Bernitsas 2011). A sizeable portion of aquatic energy therefore remains unharnessed.

Vortex Induced Vibration for Aquatic Clean Energy (VIVACE) systems provide a method of utilising these slower flow rates via a phenomenon known as vortex induced vibrations (VIV).

1.1 Vortex Induced Vibrations

A cross flow incident on a bluff body such as a cylinder generates vortexes on the objects sides. When formed irregularly these vortexes form a shear surface that oscillates behind the object: a phenomena known as vortex shedding, seen in Figure 1.

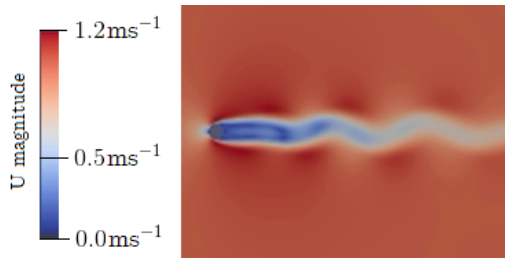


Figure 1: An example image of VIV. An Open-FOAM simulation of fluid flow around a cylinder at $Re = 100$, exhibiting a clear periodic laminar wake

These oscillating vortexes generate a perpendicular lift force on the object, which can be converted to kinetic energy. The frequency at which these vortexes oscillate is known as the Strouhal frequency

$$\omega_{St} = \frac{2\pi SrU}{D} \quad (1)$$

where U is the incident flow velocity, D is the characteristic length of the bluff body (diameter in the case of a cylinder) and Sr is the dimensionless Strouhal number, a value that relates to how turbulent flow is in the system or the flow regime. Often to help determine the flow regime of a system the ratio of inertial to viscous forces is used, also known as the Reynolds number

$$Re = \frac{UD}{\nu} \quad (2)$$

where again D is the characteristic length of the body, U is the flow velocity and ν is the kinematic viscosity of the fluid, which for water at room temperature is $1.00 \times 10^{-6} \text{ m}^2\text{s}$ (BSI Standards Limited 1999). The characteristic flow regimes for fluid incident on a cylinder were formalised in Achenbach & Heinecke (1981) and can be seen summarised in Table

Regime	Re Range
Creeping or laminar flow	$Re < 1$
Steady separated region	$3-5 < Re < 40$
Periodic laminar wake	$40 < Re < 300$
Subcritical regime	$300 < Re < 1.4 \times 10^5$
Critical regime	$1.410^5 < Re < 1 \times 10^6$
Supercritical regime	$1 \times 10^6 < Re < 5 \times 10^6$
Transcritical regime	$5 \times 10^6 < Re$

Table 1: A table displaying the division of flow regime by Reynolds number completed in Achenbach & Heinecke (1981)

Vortex shedding, and therefore VIV occur in regimes more turbulent than periodic laminar wake. It is commonly accepted that the Strouhal number for a large portion of these turbulent regimes (as large as $Re = 10^5$) has a value of 0.22 (Fey et al. 1998).

1.2 Oscillatory power and efficiency

In a VIVACE system, a cylinder is elastically suspended in the cross flow, creating a forced damped harmonic oscillator powered by the perpendicular

force described above. The systems motion is governed by the equation

$$m\ddot{y} + b\dot{y} + ky = F(t) \quad (3)$$

where m is the oscillator mass, b is the system damping coefficient, k is the spring constant of elastic supports, y is the position of the oscillator with time and $F(t)$ is the oscillatory lift force provided by VIV. This force can be described as a function of flow velocity, U , and an oscillatory lift coefficient $C_L(t)$

$$F(t) = \frac{1}{2}\rho DLU^2 C_L(t) \quad (4)$$

where L is the cylinder length, D is the cylinder diameter and other terms have their previous meaning. Being a damped harmonic oscillator, the system has a resonant frequency

$$\omega_0 = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \quad (5)$$

where again k is the spring constant of the system and m is the oscillators mass. When the driving frequency, ω_{St} , approaches ω_0 lock-in occurs and results in maximum displacement amplitude of the cylinder. Lock-in is a phenomena in which the system adjusts so that resonance takes place. This resonant behaviour suggests an optimum flow rate exists, at which maximum amplitude is observed. It can be shown (Appendix A) that the power of the oscillating cylinder in a VIVACE system is given by

$$P_{VIVACE} = \frac{bA^2\omega^2}{2} \quad (6)$$

where b is the damping coefficient of the system, ω is the frequency of the oscillator and A is the amplitude of oscillation. This amplitude dependence of power suggests that maximum generation is also achieved at an optimal flow rate for the system. One can compare P_{VIVACE} with the power of the fluid swept out by the cylinder to determine the efficiency of the system at a particular flow rate

$$P_{FLUID} = \frac{1}{2}\rho DLU^3 \quad (7)$$

$$\eta(U) = \frac{P_{VIVACE}}{P_{FLUID}} = \frac{bA^2\omega^2}{\rho DLU^3} \quad (8)$$

where all terms have their usual previous meanings.

1.3 Betz Limit

The conversion from energy in a fluid to kinetic energy is limited by Betz's law. This law is derived via the conservation of mass and momentum of fluid that is flowing through a perfect disk. Due to the geometry of any turbine system, the theoretical maximum efficiency of energy converted by a turbine from a fluid to kinetic energy is 59.3% as used in Lellis et al. (2018). This value is often used in other papers such as Lee & Bernitsas (2011) to give the efficiency of a VIVACE generator as a percentage of the Betz limit as the system can not achieve more than this theoretical maximum.

This experiment sets out to manufacture a small scale VIVACE system, to monitor its efficiency throughout parameter sweeps of mass and velocity. It is expected that the relationship between velocity and efficiency is Gaussian in nature, due to the resonant behaviour of a harmonic oscillator.

2 Designing the model VIVACE system

In order to achieve the largest power output from the system, internal power loss must be minimal. Bearings were included in the design to minimise friction between the cylinder and the supports during oscillation, and large tolerances (≈ 0.5 mm) were left surrounding the bearings inside the supports, as seen in Figure 3.

In order to accurately observe the motion of the oscillator, a brightly coloured marker was included in the design. This produces a high contrast between the marker and behind the system, allowing motion tracking software to more efficiently determine the location of the marker. The marker was polystyrene, and a lightweight bamboo dowel was used to mount it to the cylinder. These lightweight materials were chosen to maintain the balance of the cylinder.

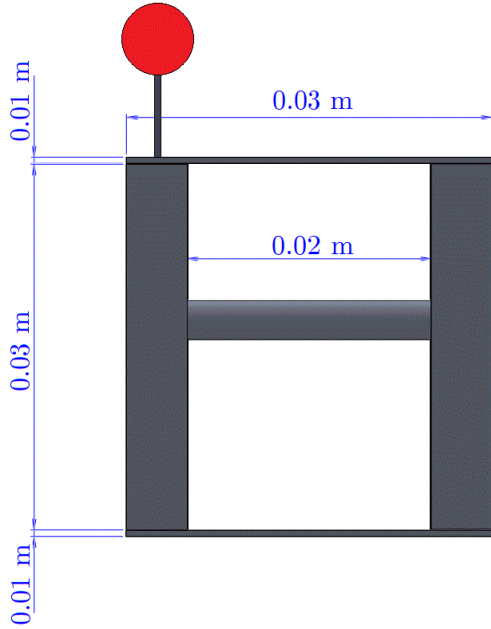


Figure 2: A schematic of the VIVACE system design from the side, showing system dimensions facing the cross flow.



Figure 3: An aerial schematic of the VIVACE model, showing the bearing/bracket mechanism.

It was necessary that the mass of the oscillator was adjustable. This was achieved by installing threaded teflon caps to the ends of the hollow cylinder. This allowed the caps to be removed between runs, and mass to be added inside the cylinder, water in this case.

The final design consisted of a spring mounted hollow cylinder with motion restricted to one degree of freedom. All parts seen in Figure 2 and Figure 3 were manufactured from aluminum.

3 Experimental method

3.1 Coefficients of the system

3.1.1 Spring constant

Firstly the spring constant k of the system was calculated. This was achieved by applying a mass to the system springs, and measuring the extension of the springs in response. This process was repeated with increasing mass. The experimental setup used can be seen in Figure 4 and masses were increased in 25/50g increments.

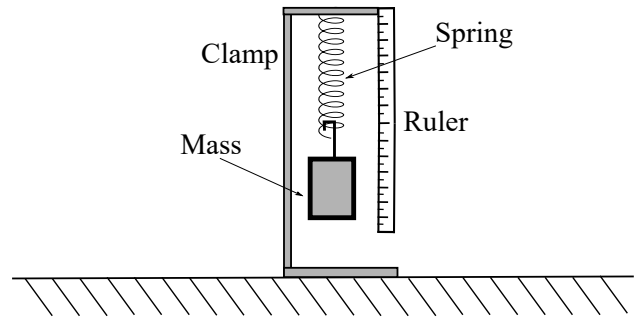


Figure 4: A diagram of the experimental setup used to determine the system spring constant.

In analysis the mass applied could be multiplied by acceleration due to gravity, g , giving the force applied to the springs. A plot of this force against spring extension was then able to be plotted and fit to a linear function. From Hooke's law it is known that the gradient of this fit is the spring constant of the system.

3.1.2 Damping coefficient

The second value to be determined is the damping coefficient of the system in water.

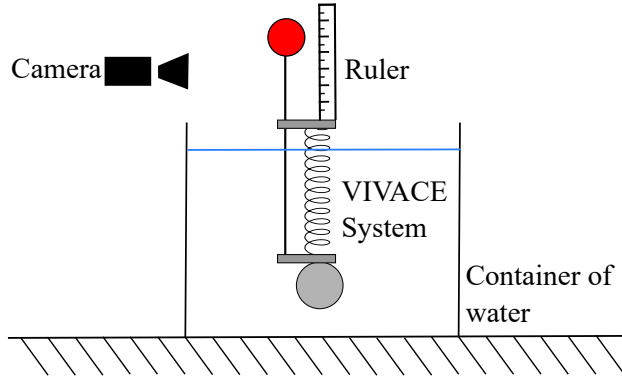


Figure 5: A diagram of the experimental setup used to determine the system's damping coefficient b in water.

The system was arranged as seen in Figure 5, with the video camera focused on the marker. The cylinder was then partially filled, weighed and the oscillator was displaced by 5cm. The motion following this displacement was recorded. Motion tracking software could be applied to the footage and a plot of the markers y-position against time could be made. The expected subsequent motion is that of light damping, where the amplitude decays exponentially as a function of the damping coefficient b

$$A \propto e^{-\frac{b}{2m}t} \quad (9)$$

where A is the oscillation amplitude and m is the mass of the oscillator. Therefore the data collected could be fit to an exponentially decaying sinusoidal function, and the coefficients of the exponential envelope used to calculate damping coefficient b . This process was repeated 10 times to allow the standard deviation in b to be calculated.

3.2 Testing of the system

The mass of the oscillator can be split into the mass of the cylinder, and the mass of auxiliary components (bearings, marker, bamboo dowel etc.). The mass cylinder was filled in 10g increments from 230-270g and at each mass the VIVACE system was placed in a flume as seen in Figure 6. The flume was filled to submerge the cylinder, and then flow rate was increased

until VIV could be observed. The water depth in the flume, D , was measured using a vernier scale depth gauge, and the volumetric flow rate was calculated by recording the time, T , taken for 300L of water to enter the flume. Multiplying the volumetric flow rate, $V = \frac{0.31}{T}$, by the cross sectional area of the flume, $A = D \times 0.305\text{m}$ gives the average flow velocity incident on the cylinder. Once these values were taken, motion of the oscillator was recorded for 25 seconds, before the valve of the flume was opened by 30° , and measurements of the same variables were taken. This process was repeated in 30° increments until VIV was no longer observed, at which point the system was removed and the cylinder mass was increased by 10g.

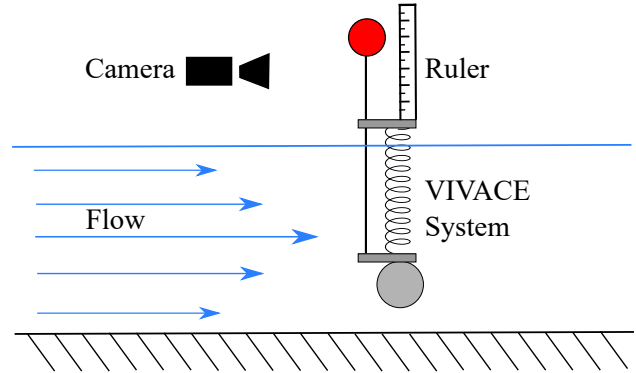


Figure 6: A diagram of the experimental setup used to monitor the oscillatory motion of the cylinder.

Motion tracking software was used to plot the y-position of the marker, the instantaneous velocity of the marker and the y-position in pixels with time. The latter two were used in the calculation of position uncertainty (Appendix B). Next by minimising chi-squared, the position of the marker was fit as a function of time. Depending on the flow regime of the system, the motion of the oscillator may exhibit a combination of frequencies. The most dominant frequencies could be identified by a fast fourier transform (FFT) of the data, and a fit of a combination of sinusoidal functions could be made. Each run was split into 10 portions of data and the fitting process was completed on each portion, allowing average am-

plitudes and frequencies to be found for each, along with an error. The values attained for A and ω could then be used in (6) to calculate the power of the oscillator in that run as well as the system efficiency.

With the above processes it was also possible to determine the dimensionless Strouhal number for the flow regime present in data collection. By making a plot of the predictably Gaussian relationship between system efficiency and U , one is able to find the flow velocity at which lock in occurs for each mass. The Strouhal frequency at this velocity is equal to the systems resonant frequency, so Eq (1) can be used to calculate the Strouhal number at each of the masses. This could be averaged across all the masses, and the standard deviation from this mean can be used as the uncertainty.

4 Results

4.1 Coefficients of the system

4.1.1 Spring constant

It was determined that the spring constant of the system k was $25.26 \pm 0.01 \text{ Nm}^{-1}$, calculated using a least-squares fit. The masses used to apply force to the springs in the system were weighed to a $\pm 0.001\text{g}$ precision using digital scales and the extension of the springs were measured to a $\pm 1 \text{ mm}$ precision using a ruler. Thus uncertainty in the extension distance of the springs is the dominant source of error in k . The uncertainty of $\pm 0.01 \text{ Nm}^{-1}$ in the value k was taken to be the standard deviation of data points from the fit seen in Figure 7.

4.1.2 Damping coefficient

The damping coefficient was found to be $b = 0.194 \pm 0.001 \text{ Nsm}^{-1}$. This was also calculated using a least squares fit of the motion of the cylinder, to an exponentially decaying sinusoidal function. As described in Section 3.1.2 the value of b was determined using the coefficients of this fit and Eq (9). Unlike the spring constant calculation, the error in this value is via the standard deviation of b over a series of 10 repeats.

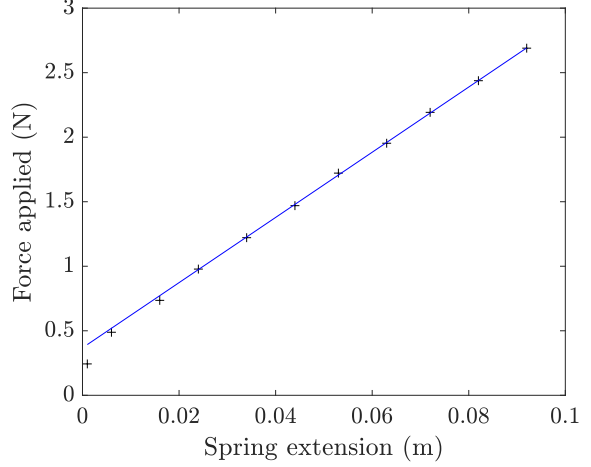


Figure 7: A plot of spring extension as a function of applied force, used to fit a linear function with a gradient equal to the spring constant of the setup.

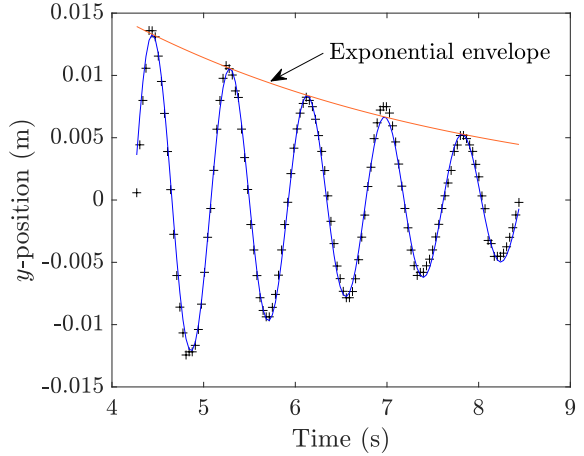


Figure 8: An example plot of the decaying sin function fit to data collected to determine the damping coefficient of the model in water.

In all 10 repetitions the motion tracking software was unable to accurately determine the position of the marker during its first oscillation. This was due to a large amount of image blur and therefore the first period was omitted from the fit. Another potential source of error arose due to the wake left by the converter. The momentum of fluid flow in the wake of the cylinder affected the motion of successive oscillations and resulted in an amplitude decay that is not perfectly exponential. This effect was most dominant in the later periods with small amplitudes, so only periods 2-6 were used in the fitting process. Figure 8 shows clearly the exponential envelope of the fit, and how later periods deviated more from the fit due to the cylinder's wake. The largest source of error in the calculation of b is because of the cylinder's wake, so future repeat of this portion of the experiment could benefit from a more massive cylinder as the momentum of fluid would have a smaller effect on its motion.

4.2 Testing of the system

4.2.1 Fitting system motion

Lock-in and oscillation was successfully achieved for all 5 masses, and the velocities tested for all masses were in the subcritical flow regime (Re to the order of 10^3). An example plot of the systems motion can be seen in Figure 9 and shows clearly an inconsistent amplitude of the marker with time. It was also suspected in Section 3.2 that the motion of the cylinder may show a combination of more than one natural frequency of the VIVACE converter. An example of a fast fourier transform made on the data can be seen in Figure 10 and shows only a single dominant mode. This was the case for all FFTs completed on data collected in this experiment and thus in subsequent fitting the system could be approximated as having only one frequency.

Each portion of the data was then fit individually to the arbitrary sin function $y(t) = A \sin \omega t$ by minimising chi-squared. An example of the data splitting can be seen in Figure 9. The coefficients A and ω of each portion were averaged, and the standard deviation from the mean was used as their error. A plot of

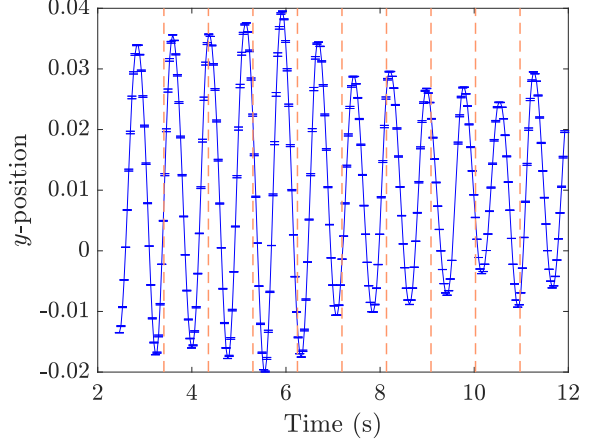


Figure 9: An example plot of oscillator position as a function of time, showing the subdivisions made for error analysis.

the averaged fit constructed from the data in Figure 9 can be seen in Figure 11.

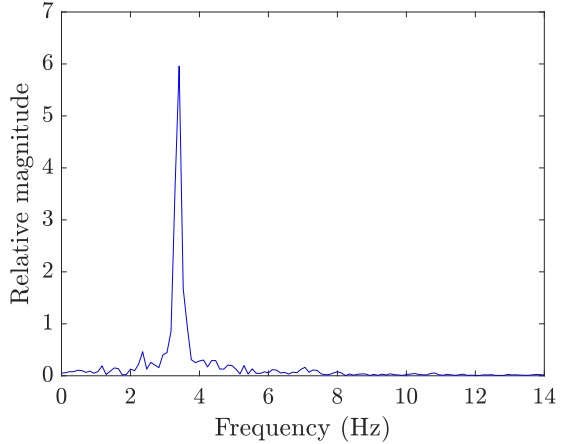


Figure 10: An example fast Fourier transform (FFT) of data collected for a single run, a plot of the relative magnitude of each frequency.

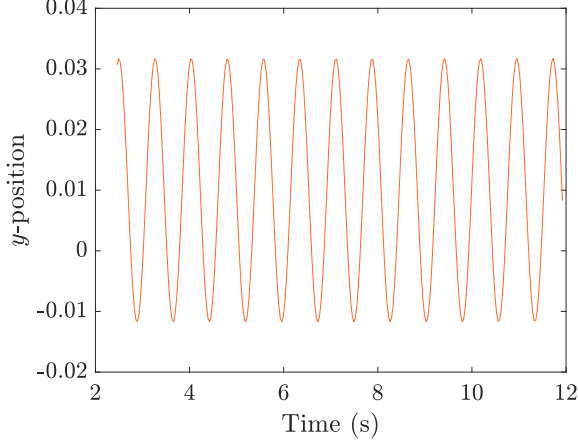


Figure 11: A plot of the averaged fit constructed from Figure 9, oscillator position as a function of time.

4.2.2 Power and efficiencies

The fitting process was applied to every run, and the amplitudes and frequencies of the averaged fits were used in Eq (6) to calculate the average power of the system at each flow velocity and the errors in damping coefficient, amplitude and frequency were propagated in quadrature. This power was used to calculate the efficiency of the system at all velocities via (8), and again errors in U and P_{VIVACE} were propagated in quadrature.

This allowed a plot of system efficiency against flow velocity to be plotted for each mass. An example of which can be seen in Figure 12 for $m = 301.8\text{g}$. For this mass, the theoretical flow velocity for peak amplitude is 0.21 ms^{-1} . The experimental peak efficiency however appears to occur at $U = 0.18\text{ ms}^{-1}$, a value smaller than the theoretical equivalent. This is perhaps due to the more turbulent flow at higher velocities resulting in a larger portion of energy being converted to rotational energy. This is enabled by the large tolerances included in the design, and is exhibited at all masses tested throughout this experiment.

Motion/big_mass-eps-converted-to.pdf

Figure 12: A plot of system efficiency against flow velocity for mass $m = 301.8\text{g}$.

The plot shows the data collected for the largest mass used, and the largest efficiency produced of 0.214 ± 0.026 . This value is comparable to other single cylinder VIVACE generators in literature. The system harnessed 70% of the 0.308 efficiency achieved by Bernitsas et al. (2009). It also sits within the range of peak efficiencies 0.196 - 0.261 harnessed in Sun et al. (2016), another single cylinder VIVACE system. The plot also confirms the Gaussian nature of the relationship between efficiency and flow velocity. Figure 12 also shows a circled data point, far from the Gaussian fit. No oscillation was recorded below a flow rate of 0.1ms^{-1} , thus the fit was forced through the point $U = 0.1, \eta = 0$, giving the appearance that the circled data point was omitted. It was however included in the fitting process.

The peak efficiencies achieved at other masses by the converter can be seen in Figure 13 and shows that over this small range of data the relationship between η and m appears to be linear. This cannot be the case for higher masses however, as the efficiency cannot exceed the Betz limit as explained in

Section 1.3, and a linear relationship would result in an efficiency above this limit at a large enough mass. For this reason, extrapolation of the efficiency for a more massive system is not possible and further research on this relationship would be beneficial. It is likely however that the efficiencies calculated in literature were collected using a more dense cylinder, of solid aluminum or steel. So if the positive correlation in Figure 13 was applicable to these larger masses, it would explain the larger value of efficiency found in Bernitsas et al. (2009).

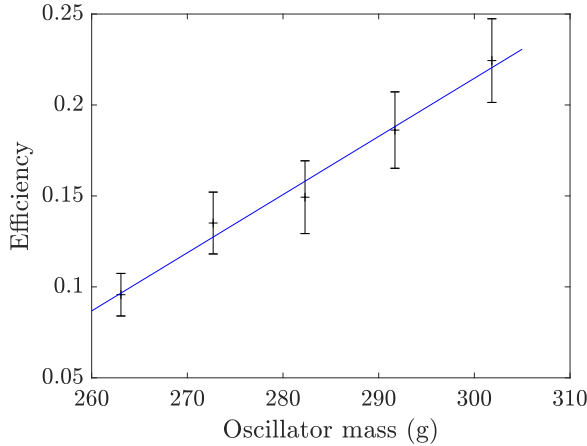


Figure 13: A plot of maximum efficiency against system mass, with an approximate linear fit.

The error in the peak efficiency values is largely due to the variation in amplitudes averaged to calculate the power. The marker used to track the cylinder adds a mass to only one side of the oscillator and the large tolerances included in the design allow the system cylinder to rotate within the supports. Because of this, the position of the marker does not always accurately represent the mean displacement of the cylinder along its full length. This rotation is a likely explanation for the seemingly chaotic variation in amplitude, thus in a repeat of this experiment it would be beneficial to restrict cylinder rotation. To a lesser extent, error in the efficiencies is also caused by uncertainty in flow velocity. The measurement of time taken for 300L to enter the system contributed a 0.5s error due to human reaction times. If this pro-

cess were to be repeated it would be beneficial to use a less erroneous technique of measuring flow velocity incident on the cylinder, perhaps a digital flow meter.

As well as peak efficiency, the largest average efficiency of the generator was also achieved at $m = 301.8\text{g}$, and had a value of 0.153 ± 0.052 . This value is just below the range of peak efficiencies achieved by Sun et al. (2016), and is almost exactly half the peak efficiency harvested in Bernitsas et al. (2009). At the small scale of the model VIVACE converter and slow flow rates of the flume used, this equates to a power output of $3.10 \pm 1.11\text{ mW}$. A value small on the scale of large hydro kinetic implementations, however a larger frontal area of the cylinder would increase this value. Thus, assuming a 90% efficiency in the conversion from kinetic to electrical energy, the power needs of the average house (3,638 kWh a year from The Energy Saving Trust (2011)) could be met by a single full size VIVACE converter ($\approx 1\text{m}$ radius, 6m length) in average current flow of 2 ms^{-1} . This value would likely decrease if the density of the cylinder used was larger than in this experiment, dependent on the relation between efficiency and oscillator mass.

4.2.3 Strouhal number

By fitting a Gaussian function to efficiency against flow velocity for all 5 masses (seen in Figure 12) the resonance peak of the system could be determined. As explained previously this peak occurs at lock-in, so the frequency of the oscillator at the peak is equal to the driving frequency of VIV. By substituting the flow velocity of the peak efficiencies and their frequencies into (1), it was possible to calculate the dimensionless Strouhal number. By this process the Strouhal number was determined to be 0.218 ± 0.020 . This value is within one error of the commonly accepted value found in Mendez et al. (2017) and Fey et al. (1998) of 0.22 over large range of Reynolds numbers for flow regimes with VIV.

5 Conclusions

Vortex induced vibrations is a phenomena in which fluid flow incident on a bluff body generates an vortexes along the sides of the object. This creates a shear surface that applies an oscillating force to the object, perpendicular to the direction of flow. A VIVACE generator aims to generate hydro-kinetic energy via the oscillation of a cylinder as a result of VIV. This experiment set out to design a VIVACE generator and investigate the effect of flow rate and oscillator mass on the converter's efficiency. Firstly the spring constant of the system was determined to be $25.26 \pm 0.01 \text{ Nm}^{-1}$ by the classic Hooke's law experiment. The damping coefficient of the system in water was next calculated to be $0.194 \pm 0.001 \text{ Nsm}^{-1}$ in still water. The y -position of the cylinder was plotted as a function of time by applying motion tracking software to footage of a brightly coloured marker attached to the oscillator. The final manufactured system achieved lock in for a range of velocities at a range of masses and oscillation consisted of only a single dominated frequency as demonstrated by a fast fourier transform of the data. Thus fits of single frequency sinusoidal functions were made at each velocity and mass, and the coefficients for amplitude and frequency were used to calculate the average efficiency. The peak efficiencies achieved were between 0.096 ± 0.022 at the lowest mass of $m = 263.1 \text{ g}$ and 0.215 ± 0.022 at the largest mass of $m = 301.8 \text{ g}$. The upper value of which lies within the range of 0.196 - 0.261 achieved by another single cylinder VIVACE system in Sun et al. (2016). This maximum also constitutes 36% of the Betz limit, a theoretical ceiling for kinetic energy gathered from fluid flow. Throughout this process it was also confirmed that the Strouhal number for the system averaged 0.218 ± 0.020 , which is in agreement with the generally accepted value of 0.22 in Mendez et al. (2017) and Fey et al. (1998) for flow in regimes that generate VIV. In future experiments a more accurate measurement of efficiency could be determined, by using a cylinder that had restricted rotation. This would result in a significantly smaller standard deviation in amplitude displacement averages, and provide a much smaller margin of error in efficiency. Future experiments on

the effect of larger masses on system efficiencies would also be beneficial as the data collected throughout this experiment suggests a linear relationship. This is not possible however, as the efficiency cannot exceed the Betz limit of 59.8%, and a positively correlated linear fit suggests an efficiency greater than this at a larger mass.

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A Derivation of power output

The time averaged power of an oscillator is given by

$$P = \frac{1}{T_{osc}} \int_0^{T_{osc}} F(t) \dot{y} dt$$

where T_{osc} is the period of the oscillators motion, $F(t)$ is the force acting upon the cylinder and $\dot{y}$ is the velocity of the cylinder. One can substitute the equation of motion for a VIVACE converter (3) to give

$$P = \frac{1}{T_{osc}} \int_0^{T_{osc}} (m\ddot{y} + b\dot{y} + ky) \dot{y} dt$$

If we assume that position y as a function of time is $A \sin \omega t$. From this substitution the terms containing mass, m , and spring constant, k , disappear as they average 0 over the integral bounds. This gives

$$P = \frac{1}{T_{osc}} \int_0^{T_{osc}} bA^2\omega^2 \cos^2(\omega t) dt$$

which over the integral bounds gives the final equation (6)

$$P = \frac{bA^2\omega^2}{2}$$

giving a result for the power of the converters motion to be dependent only on the damping coefficient b , the oscillator amplitude A and the frequency of the oscillations ω .

B Error calculation of oscillator position

The source of error in the y -position of the marker is from camera blur at higher velocities. The motion tracking software used throughout the analysis of this experiment was unable, however, to output the uncertainty in y position of the marker. For this reason it was necessary in error calculation, that the velocity be the main factor used to calculate position uncertainty. Footage of the marker at a number of velocities was analysed and a plot of image blur b against marker velocity v was made. A linear fit between these two variables was made, as seen in Figure 14.

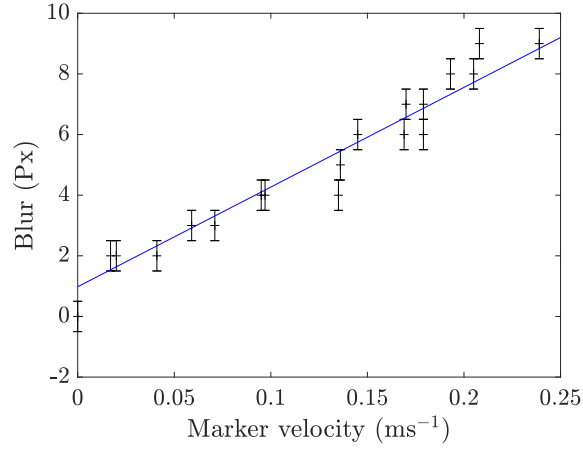


Figure 14: The plot of image blur as a function of motion velocity used for error calculation. All points possess a ± 0.5 px precision error.

A least squares fit was made to the plot above, due to all points having the same error in blur. The tracking software was (similar to position) unable to output an uncertainty in marker velocity. The function was calculated to be $b = 32.88v + 0.98$. This function was then applied to the instantaneous velocity at each data point, giving all points a blur value, which was interpreted as the uncertainty in y -position. This allows a more accurate chi-squared fit to be made in the analysis of motion, instead of the less accurate least-squares fit.